

SOFTWARE SPECIFICATION DOCUMENT

Hospital Management System (HMS)

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1. Introduction

1.1 Purpose

The Hospital Management System (HMS) is developed to automate and manage hospital operations efficiently. The system centralizes patient information, appointments, billing, pharmacy, laboratory services, and administrative tasks to improve healthcare service delivery.

1.2 Scope

The system provides:

- Patient Registration
 - Doctor Management
 - Appointment Scheduling
 - Electronic Medical Records
 - Laboratory Management
 - Pharmacy Management
 - Billing and Payments
 - Inpatient and Outpatient Management
 - Staff Management
 - Reports and Analytics
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2. System Overview

The system is used by:

- Administrator
- Doctors
- Nurses
- Receptionists
- Laboratory Staff
- Pharmacists
- Accountants

3. Functional Requirements

3.1 User Management

- Login
- Logout
- Password Management
- User Roles
- Access Control

3.2 Patient Management

- Register Patient
- Update Patient Information
- Search Patient
- View Medical History
- Upload Documents

3.3 Doctor Management

- Add Doctor
- Update Doctor Details
- Department Assignment
- Schedule Management

3.4 Appointment Management

- Book Appointment
- Cancel Appointment
- Reschedule Appointment
- Appointment Status Tracking

3.5 Electronic Medical Records

- Patient Diagnosis
- Prescriptions
- Treatment History
- Medical Reports

3.6 Laboratory Management

- Test Requests
- Sample Collection
- Result Entry
- Report Generation

3.7 Pharmacy Management

- Medicine Inventory
- Prescription Processing
- Stock Management
- Expiry Monitoring

3.8 Billing System

- Consultation Charges
- Laboratory Charges
- Pharmacy Charges
- Admission Charges
- Invoice Generation
- Payment Recording

3.9 Staff Management

- Employee Registration
- Attendance
- Department Assignment

- Leave Records

3.10 Reports

- Patient Reports
 - Appointment Reports
 - Revenue Reports
 - Pharmacy Reports
 - Laboratory Reports
 - Staff Reports
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4. Non-Functional Requirements

Performance

- Supports multiple concurrent users
- Fast response time
- Optimized database performance

Security

- Secure Login
- Password Encryption
- Role-Based Access Control
- Audit Logs
- Automatic Session Timeout

Reliability

- Daily Database Backup
- High Availability
- Data Recovery Support

Usability

- User-Friendly Interface

- Responsive Design
- Easy Navigation

5. System Architecture

Presentation Layer

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Application Layer

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Business Logic Layer

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Database Layer

6. Database Modules

- Users
- Roles
- Patients
- Doctors
- Departments
- Appointments
- Admissions
- Medical Records
- Laboratory Tests
- Pharmacy
- Billing
- Payments
- Employees

7. User Interface Specifications

Login Page

- Username
- Password
- Login Button

Dashboard

- Total Patients
- Today's Appointments
- Revenue Summary
- Laboratory Requests
- Pharmacy Alerts

Patient Screen

- Add Patient
- Edit Patient
- Search Patient
- View History

Appointment Screen

- Schedule Appointment
- Calendar View
- Doctor Selection

Billing Screen

- Generate Bill
 - Receive Payment
 - Print Receipt
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8. Hardware Requirements

Minimum:

- Intel Core i5 Processor
- 8 GB RAM
- 256 GB SSD
- Printer
- LAN Connection

Recommended:

- Intel Core i7
 - 16 GB RAM
 - SSD Storage
 - Backup Server
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9. Software Requirements

Server

- Windows Server or Linux
- Apache/Nginx/IIS
- PHP, Java, .NET, or Node.js
- MySQL, PostgreSQL, SQL Server, or Oracle

Client

- Windows 10/11
 - Google Chrome
 - Microsoft Edge
 - Mozilla Firefox
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10. Backup and Recovery

- Automatic Daily Backup
 - Weekly Full Backup
 - Disaster Recovery Plan
 - Audit Trail
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11. Benefits

- Faster Patient Registration
 - Improved Appointment Management
 - Accurate Billing
 - Better Inventory Control
 - Reduced Paperwork
 - Secure Medical Records
 - Enhanced Patient Care
 - Better Decision Making through Reports
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12. Future Enhancements

- Mobile Application
- Patient Portal
- SMS & Email Notifications
- Telemedicine
- Insurance Integration
- AI-Based Decision Support
- Biometric Authentication
- Cloud Deployment

